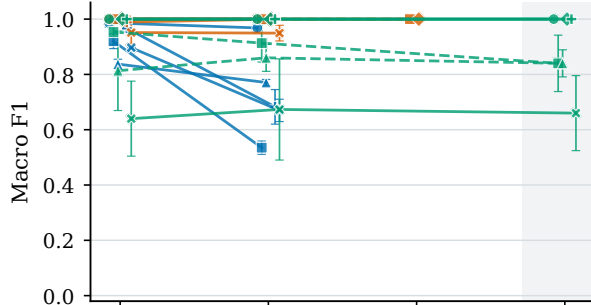


# Scaling by tier across six models

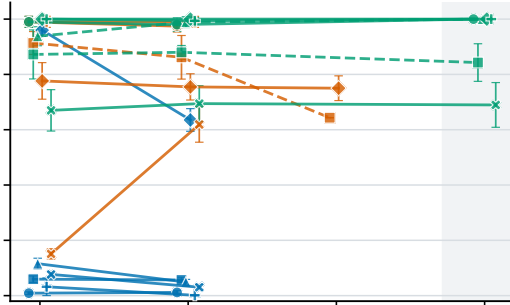
LLM CodeAct RLM

● Qwen 3.5 ■ \*DeepSeek V4 Flash ▲ \*GLM 5.2 ◆ Gemini 3.7 Flash + GPT-5 mini ✖ Claude Haiku 4.5

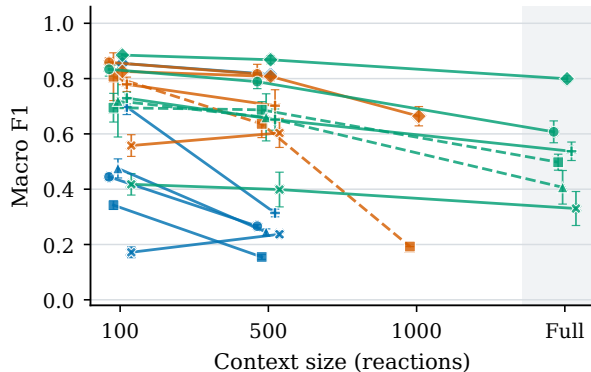
(a) Tier 1: Structural lookup ( $n=10$ )



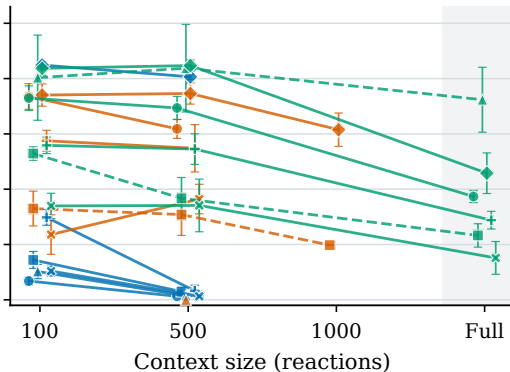
(b) Tier 2: Property aggregation ( $n=20$ )



(c) Tier 3: Reaction-level reasoning ( $n=35$ )



(d) Tier 4: Relational reasoning ( $n=35$ )



Terminal failed jobs score zero; unresolved jobs are excluded. Small offsets improve visibility; dashed curves and \* labels are provisional.